

Q1/ Define the following:

1. Atmospheric windows. 2. Passive sensors. 3. The swath. 4. Active sensors. 5. Mie scattering. 6. Rayleigh scattering. 7. Resolution cell. 8. Nadir point. 9. The wavelength. 10. The primary colours.

Q2/ Write the elements that comprise the remote sensing process from beginning to end.

Q3/ Fill in the following blanks:

1. The first requirement for remote sensing is to have an **energy source** to illuminate the target.
 2. **Nonselective** scattering occurs when the particles are much larger than the wavelength of the radiation.
 3. The part of the electromagnetic spectrum that the Earth typically emits as heat is **Thermal IR**.
 4. The visible wavelengths cover a range from approximately **0.4 to 0.7** μm .
 5. **Specular** reflection occurs on a smooth surface.
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Q4/ What are the main atmospheric constituents that absorb radiation? Explain them.

Q5/ How does chlorophyll interact with different wavelengths of light, specifically red, blue, green, and near-infrared?

Q6/ What are the Advantages of active sensors?

Q7/ What are the differences between images and photographs.

Q8/ Why in near polar orbits, areas at high latitudes will be imaged more frequently than the equatorial zone?

Q9/ If a remote sensing image displays data at full resolution, and each pixel on that image represents an area of 40 meters x 40 meters on the ground, what can be concluded about the sensor's spatial resolution?

Q10/ Choose the correct answer:

1. What is the first requirement for remote sensing to occur?

- A. A satellite
- B. A camera
- C. An energy source
- D. A computer

2. Which of the following is NOT one of the seven elements of the remote sensing process?

- A. Data Encryption

- B. Application
- C. Energy Source
- D. Interaction with the Target

3. What portion of the electromagnetic spectrum do our eyes detect?

- A. Infrared
- B. Visible
- C. Ultraviolet
- D. Microwave

4. What type of scattering is caused by particles much larger than the wavelength of radiation?

- A. Rayleigh
- B. Mie
- C. Nonselective
- D. Reflective

5. What term describes the range of wavelengths where electromagnetic energy is not absorbed by the atmosphere?

- A. Energy bands
- B. Wavelength channels
- C. Atmospheric windows
- D. Scattering gaps

6. Which part of the electromagnetic spectrum is typically emitted by the Earth as heat?

- A. Visible
- B. Reflected IR
- C. Thermal IR
- D. Ultraviolet

7. What is the difference between passive and active remote sensing systems?

- A. Passive systems emit their own energy
- B. Active systems rely on the sun
- C. Active systems emit their own energy
- D. Passive systems work at night only

8. What is a pixel in the context of remote sensing images?

- A. A camera lens
- B. A measuring tool
- C. The smallest unit of an image
- D. A type of wave

9. What kind of orbit allows satellites to view the same portion of the Earth at all times?

- A. Polar orbit
- B. Circular orbit
- C. Geostationary orbit
- D. Elliptical orbit

10. What does spatial resolution refer to?

- A. The distance a satellite travels in an hour
- B. The brightness of the image
- C. The smallest feature that can be detected
- D. The color depth of an image

11. Which field is perpendicular to the direction of travel in electromagnetic radiation?

- A. Sound
- B. Electric
- C. Gamma
- D. Alpha

12. Which unit is commonly used to measure wavelength?

- A. Kilogram
- B. Joule
- C. Nanometer
- D. Second

13. Which wavelength range is typical for visible light?

- A. 0.1 to 0.3 μm
- B. 0.4 to 0.7 μm
- C. 1.0 to 2.5 μm
- D. 10 to 100 μm

14. Which color has the shortest wavelength in the visible spectrum?

- A. Red
- B. Green

- C. Blue
- D. Violet

15. What is the role of chlorophyll in remote sensing?

- A. It reflects red light
- B. It absorbs green light
- C. It reflects near-infrared and green light
- D. It emits thermal radiation

16. Which interaction results in energy passing through the target?

- A. Absorption
- B. Emission
- C. Reflection
- D. Transmission

17. What type of reflection occurs on a smooth surface?

- A. Diffuse
- B. Rough
- C. Specular
- D. Scattering

18. In passive sensing, what is the primary energy source?

- A. The sensor itself
- B. A satellite
- C. The sun

D. Earth's heat

19. What distinguishes an image from a photograph in remote sensing?

- A. Image uses film, photograph is digital
- B. Photograph is any visual, image is always digital
- C. Photograph is film-based, image can be any pictorial data
- D. They are the same

20. What is a channel in remote sensing?

- A. A tool for measuring radiation
- B. A sensor movement path
- C. A narrow range of wavelength data
- D. The physical satellite path

21. What is the main advantage of active sensors over passive sensors?

- A. Higher resolution
- B. Operate only during daytime
- C. Can operate regardless of time or season
- D. Cheaper to operate

22. What is the name of the point on the Earth's surface directly below a satellite?

- A. Swath
- B. Zenith
- C. Nadir
- D. Apex

23. Which of the following refers to a satellite that appears fixed relative to Earth?

- A. Polar orbit
- B. Geostationary orbit
- C. Elliptical orbit
- D. Retrograde orbit

24. Which orbit allows satellites to cover most of Earth over time due to the planet's rotation?

- A. Geostationary
- B. Equatorial
- C. Near-polar
- D. Elliptical

25. What is a satellite swath?

- A. The altitude of a satellite
- B. The angle of inclination
- C. The width of the ground area imaged by the satellite
- D. The rotation speed of the Earth

26. How is spatial resolution defined?

- A. Brightness of the image
- B. Amount of memory used
- C. The smallest object detectable in an image
- D. Distance covered by a satellite in a day

27. What is the typical range of spatial resolution in commercial satellite images?

- A. Centimeters to meters
- B. Meters to kilometers
- C. Kilometers to miles
- D. Micrometers to millimeters

28. What does pixel size represent in remote sensing imagery?

- A. A digital storage unit
- B. A wavelength measure
- C. Ground area represented by one pixel
- D. Image contrast level

29. Which sensor placement is best for detailed ground-level information?

- A. Space-based
- B. Aerial
- C. Ground-based
- D. Balloon

30. Which is a characteristic of aerial platforms?

- A. Most expensive
- B. Provide continuous imaging over time
- C. Provide flexible and detailed data collection
- D. Used mainly for navigation

31. What kind of satellites are typically placed in geostationary orbits?

- A. Communication
- B. Military
- C. Meteorological
- D. All of the above

32. What happens to revisit time when using steerable sensors?

- A. Increases
- B. Remains unchanged
- C. Decreases
- D. Becomes irregular

33. What term describes images that show only large features?

- A. Fine resolution
- B. High scale
- C. Coarse resolution
- D. Narrow swath

34. What does a scale of 1:100,000 mean?

- A. 1 cm on image equals 1,000 m on ground
- B. 1 km on image equals 1 cm on ground
- C. 1 m on image equals 1 mm on ground
- D. 1 cm on image equals 1 m on ground

35. What is the term for combining channels with primary colors to display images?

- A. RGB mapping

B. Color synthesis

C. False color processing

D. Digital encoding

36. Which of the following absorbs ultraviolet radiation in the atmosphere?

A. CO₂

B. Ozone

C. Water vapor

D. Dust

37. What are the two main types of infrared radiation used in remote sensing?

A. Ultraviolet and visible

B. Visible and microwave

C. Reflected IR and Thermal IR

D. X-rays and gamma rays

38. What type of scattering affects shorter wavelengths most?

A. Mie

B. Rayleigh

C. Nonselective

D. Thermal

39. Which gas absorbs both longwave infrared and shortwave microwave radiation?

A. Carbon dioxide

B. Ozone

C. Water vapor

D. Methane

40. What kind of image results from displaying a single channel in grayscale?

A. Thermal image

B. Black and white image

C. Color image

D. UV image